

Name: Answers

Exam Score: _____

1. (14 pt) For what vectors $b \in \mathbb{R}^3$ is $Ax = b$ consistent when

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & -3 \\ 0 & 1 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 1 & -1 & b_1 \\ 2 & 1 & -3 & b_2 \\ 0 & 1 & 1 & b_3 \end{array} \right] \xrightarrow{-2R_1+R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & b_1 \\ 0 & -1 & -1 & b_2 - 2b_1 \\ 0 & 1 & 1 & b_3 \end{array} \right]$$

$$\xrightarrow{R_2+R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & -1 & b_1 \\ 0 & -1 & -1 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 + b_2 - 2b_1 \end{array} \right]$$

$Ax = b$ is consistent if and only if $b_3 + b_2 - 2b_1 = 0$

2. (12 pt) Give the solution set of $Ax = 0$ in parametric vector form when $A = \begin{bmatrix} 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & -1 \\ 1 & 2 & 3 & 3 \end{bmatrix}$.

$$A \xrightarrow{\substack{-R_1+R_2 \rightarrow R_2 \\ -R_1+R_3 \rightarrow R_3}} \begin{bmatrix} 1 & 2 & 2 & 1 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{\substack{2R_2+R_1 \rightarrow R_1 \\ R_2+R_3 \rightarrow R_3}} \begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{-R_2} \begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ RREF of } A$$

$$\Rightarrow \text{Solution set of } Ax=0 \text{ is } \begin{cases} x_1 + 2x_2 - 3x_4 = 0 \\ x_3 + 2x_4 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -2x_2 + 3x_4 \\ x_3 = -2x_4 \end{cases}$$

Parametric Vector Form

$$x \in \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2x_2 + 3x_4 \\ x_2 \\ -2x_4 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 3 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \quad x_2, x_4 \in \mathbb{R}$$

3. (24 pt) For the following parts let $[A|b] = [a_1 \ a_2 \ a_3 \ | \ b] = \left[\begin{array}{ccc|c} 1 & -1 & 0 & -3 \\ -1 & 2 & 2 & 7 \\ 0 & 2 & 5 & 9 \\ 1 & -1 & -1 & -4 \end{array} \right] \sim B = \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$

a) Show that b can be written as a linear combination of the vectors a_1, a_2, a_3 .

$$b = -a_1 + 2a_2 + a_3 = -\begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} + 2\begin{bmatrix} -1 \\ 2 \\ 2 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 5 \\ -4 \end{bmatrix}$$

b) Are the vectors a_1, a_2, a_3 independent? Explain.

Yes, because there is a pivot in each column of A

c) If $T(x) = Ax$, is T one-to-one? Explain.

Yes, because the columns of A are independent by b)

d) If $T(x) = Ax$, is T onto? Explain.

No, because there is not a pivot in each row of A

4. (14 pt) Let A be a 3×3 matrix, and suppose that

$$A \xrightarrow{2R_1+R_2 \rightarrow R_2} B \xrightarrow{R_2 \leftrightarrow R_3} C \xrightarrow{-3R_2+R_3 \rightarrow R_3} D \xrightarrow{2R_1} I_3$$

Explain why A is invertible and find A^{-1} .

A is invertible because its rref is I_3

$$I_3 \xrightarrow{2R_1+R_2 \rightarrow R_2} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \xrightarrow{-3R_2+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -3 \end{bmatrix}$$

$$\xrightarrow{2R_1} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -3 \end{bmatrix} = A^{-1}$$

5. (12 pt) Find an LU factorization of the matrix A with L unit lower triangular and U upper triangular, and where

$$A = \begin{bmatrix} 2 & -4 & -2 & 0 \\ -2 & 1 & 3 & 1 \\ 6 & -6 & -9 & -1 \end{bmatrix} \xrightarrow{R_1+R_2 \rightarrow R_2} \begin{bmatrix} 2 & -4 & -2 & 0 \\ 0 & -3 & 1 & 1 \\ 6 & -6 & -9 & -1 \end{bmatrix} \xrightarrow{-3R_1+R_3 \rightarrow R_3} \begin{bmatrix} 2 & -4 & -2 & 0 \\ 0 & -3 & 1 & 1 \\ 0 & 6 & -3 & -1 \end{bmatrix} = B$$

Find the factorization using the information above and one of the algorithms covered in class.

$$\downarrow \frac{1}{2}$$

$$\begin{bmatrix} 1 \\ -\frac{1}{2} \\ 3 \end{bmatrix}$$

$$\downarrow \frac{1}{-3}$$

$$\begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

$$\Rightarrow L = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}$$

$$B \xrightarrow{2R_2+R_3 \rightarrow R_3} U = \begin{bmatrix} 2 & -4 & -2 & 0 \\ 0 & -3 & 1 & 1 \\ 0 & 0 & -1 & -1 \end{bmatrix}$$

Alt. derivation of L

$$I_3 \xrightarrow{-2R_2+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \xrightarrow{3R_1+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} \xrightarrow{-R_1+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} = L$$

6. (4 pt) The linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ first performs a shear that leaves $\mathbf{e}_1$ unchanged and transforms $\mathbf{e}_2$ into $\mathbf{e}_1 + 2\mathbf{e}_2$, and which then reflects points in the plane through the line $y = -x$. Circle the standard matrix for the linear transformation T .

(i) $\begin{bmatrix} 0 & -1 \\ -1 & -2 \end{bmatrix}$ (ii) $\begin{bmatrix} -2 & -1 \\ -1 & 0 \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ (iv) $\begin{bmatrix} 0 & -2 \\ -1 & -1 \end{bmatrix}$ (v) $\begin{bmatrix} -1 & -1 \\ -2 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ -1 & -1 \end{bmatrix}$

7. (4 pt) Circle the 3×3 matrix that produces the composite 2D transformation, using homogeneous coordinates, that first translates by $(3, 1)$ and then rotates 45° about the origin.

(i) $\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 3 \\ \sqrt{2}/2 & \sqrt{2}/2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ (ii) $\begin{bmatrix} 1/2 & -\sqrt{3}/2 & 3 \\ \sqrt{3}/2 & 1/2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ (iii) $\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & \sqrt{2} \\ \sqrt{2}/2 & \sqrt{2}/2 & 2\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix}$ (iv) $\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 1 \\ \sqrt{2}/2 & \sqrt{2}/2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$

8. (4 pt) Circle any values of h for which A is not invertible where

$A = \begin{bmatrix} 0 & 2 & 2 \\ 1 & 1 & 0 \\ 2 & 0 & h \end{bmatrix}$

$\det A = h \begin{vmatrix} 0 & 2 \\ 1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix}$
 $= -2h - 4 = 0$
 $\Rightarrow h = -2$

(i) $\frac{1}{2}$ (ii) -2 (iii) -4 (iv) 2 (v) $-\frac{1}{4}$

9. (8 pt) For the following parts let $A^{-1} = \begin{bmatrix} -2 & 3 \\ -1 & 2 \end{bmatrix}$, and let $B^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

- a) Circle the matrix that equals $(AB)^{-1} = B^{-1}A^{-1}$

(i) $\begin{bmatrix} -4 & 7 \\ -10 & 17 \end{bmatrix}$ (ii) $\begin{bmatrix} 17/2 & -7/2 \\ 5 & -2 \end{bmatrix}$ (iii) $\begin{bmatrix} 7 & 8 \\ 5 & 6 \end{bmatrix}$ (iv) $\begin{bmatrix} 3 & -4 \\ -7/2 & 7/2 \end{bmatrix}$ (v) $\begin{bmatrix} -4 & 7 \\ 17 & -10 \end{bmatrix}$

- b) Circle that solution for the system $Ax = \begin{bmatrix} -2 \\ 3 \end{bmatrix} \Rightarrow x = A^{-1} \begin{bmatrix} -2 \\ 3 \end{bmatrix}$

(i) $\begin{bmatrix} 7 \\ -9/2 \end{bmatrix}$ (ii) $\begin{bmatrix} 4 \\ 6 \end{bmatrix}$ (iii) $\begin{bmatrix} 4 \\ 18 \end{bmatrix}$ (iv) $\begin{bmatrix} 6 \\ 4 \end{bmatrix}$ (v) $\begin{bmatrix} 13 \\ 8 \end{bmatrix}$

10. (4 pt) Let A be a 3×3 matrix, and suppose that $\det(C) = 5$ and that

$A \xrightarrow{2R_1} B \xrightarrow{R_2 \leftrightarrow R_3} C \xrightarrow{-3R_2 + R_3 \rightarrow R_3} D$

then give the values of following

(i) $\det(A) = -5/2$ (ii) $\det(D) = 5$